Lab 5 - Modelling I

Kathrine Marattil

2026-04-05

Introduction

In this lab you'll start your practice of statistical modelling. You'll fit models, interpret model output, and make decisions about your data and research question based on the model results.

Guidelines

Your plots should include an informative title, axes should be labeled, and careful consideration should be given to aesthetic choices.

In addition, the code should all the code should be be able to be read (not run off the page) when you render to PDF. Make sure that is the case, and add line breaks where the code is running off the page.

> **Important**
>
> Render your document. If your code is running off the page for a given question such that we can't see your entire code, we will not evaluate any of the code for that question. The question will automatically receive a 0. This is something you can and should verify before you turn in your work.

You should have at least 3 commits with meaningful commit messages by the end of the assignment.

Additionally, if you're using functions that are not introduced in the course materials, you must cite your sources.

> **Important**
>
> Failure to cite outside resources used, including Large Language Models like Chat GPT, is a violation of the Duke Community Standard and will be treated as such.

Part 1 - Smoking during pregnancy

```r
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.1     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.2     v tibble    3.3.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.1
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becom
```

```r
library(openintro)
```

```
Loading required package: airports
Loading required package: cherryblossom
Loading required package: usdata
```

```r
library(broom)
library(ggplot2)
```

Question 1

We are interested in the impact of smoking during pregnancy. Since it is not possible to run a randomized controlled experiment to investigate this impact, we will instead use a data set has been of interest to medical researchers who are studying the relation between habits and practices of expectant mothers and the birth of their children. This is a random sample of 1,000 cases from a data set released in 2014 by the state of North Carolina. The data set is called `births14`, and it is included in the **openintro** package you loaded at the beginning of the assignment.

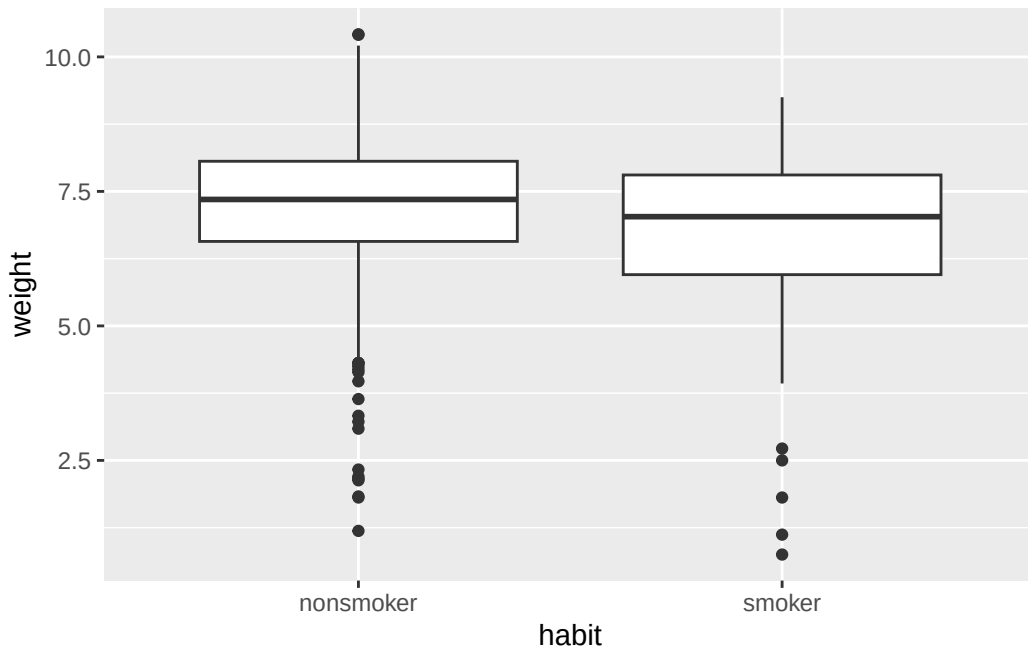
a. Create a version of the `births14` data set dropping observations where there are `NA`s for `habit`. You can call this version `births14_habitgiven`.

```
births14_habitgiven <- births14 |>
  filter(!is.na(habit))
births14_habitgiven
```

```
# A tibble: 981 x 13
    fage  mage mature     weeks premie visits gained weight lowbirthweight sex
   <int> <dbl> <chr>      <dbl> <chr>   <dbl>  <dbl>  <dbl> <chr>          <chr>
 1    34    34 younger m~    37 full ~     14     28   6.96 not low        male
 2    36    31 younger m~    41 full ~     12     41   8.86 not low        fema~
 3    37    36 mature mom    37 full ~     10     28   7.51 not low        fema~
 4    NA    16 younger m~    38 full ~     NA     29   6.19 not low        male
 5    32    31 younger m~    36 premie     12     48   6.75 not low        fema~
 6    32    26 younger m~    39 full ~     14     45   6.69 not low        fema~
 7    37    36 mature mom    36 premie     10     20   6.13 not low        fema~
 8    29    24 younger m~    40 full ~     13     65   6.74 not low        male
 9    30    32 younger m~    39 full ~     15     25   8.94 not low        fema~
10    29    26 younger m~    39 full ~     11     22   9.12 not low        male
# i 971 more rows
# i 3 more variables: habit <chr>, marital <chr>, whitemom <chr>
```

b. Plotting the data is a useful first step because it helps us quickly visualize trends, identify strong associations, and develop research questions. Create an appropriate plot displaying the relationship between `weight` and `habit`. In 2-3 sentences, discuss the relationships observed.

```
ggplot(births14_habitgiven,aes(x= habit, y = weight)) +
  geom_boxplot()
```



***

- **data for smokers shows a left skew**
- **all of the habit data has a great amount of outliers on the lower side of the wealth**

c. Now, fit a linear model that predicts `weight` from `habit`. Provide the tidy summary output below.

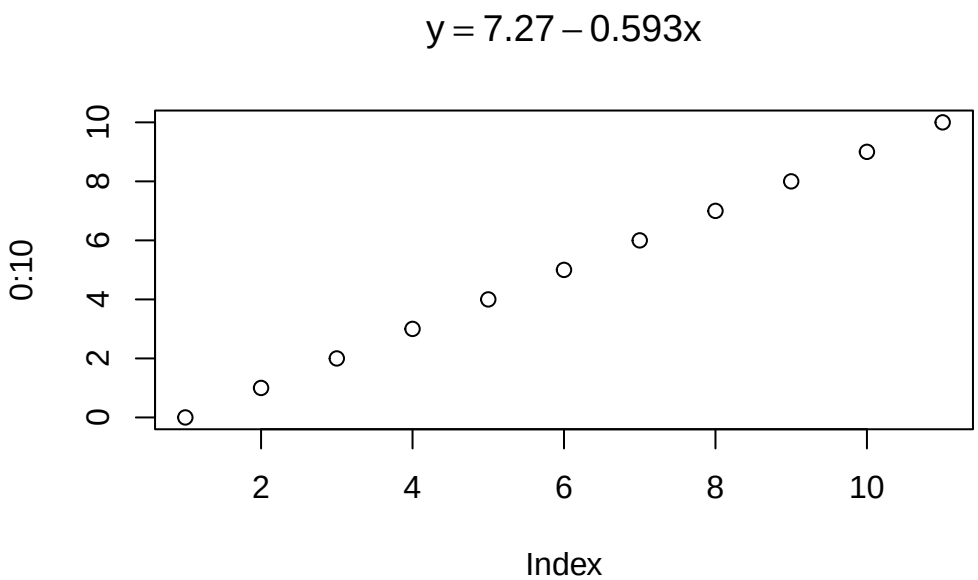
```
#linear model
linear_model<- lm (weight ~ habit, data = births14_habitgiven)

#tidy summary
tidy(linear_model, conf.int = FALSE, conf.level = 0.95)
```

```
# A tibble: 2 x 5
  term        estimate std.error statistic    p.value
  <chr>          <dbl>     <dbl>     <dbl>      <dbl>
1 (Intercept)    7.27     0.0435    167.   0
2 habitsmoker   -0.593    0.128      -4.65 0.00000382
```

d. Write the estimated least squares regression line below using proper notation.

$\hat{y}$ = 7.270 - 0.593*x

```r
plot(0:10, main = expression(y ==7.270  - 0.593*x))
```



> **Tip**
>
> If you need to type an equation using proper notation, type your answers in-between `$$` and `$$`. You may use `\hat{example}` to put a hat on a character.

Question 2

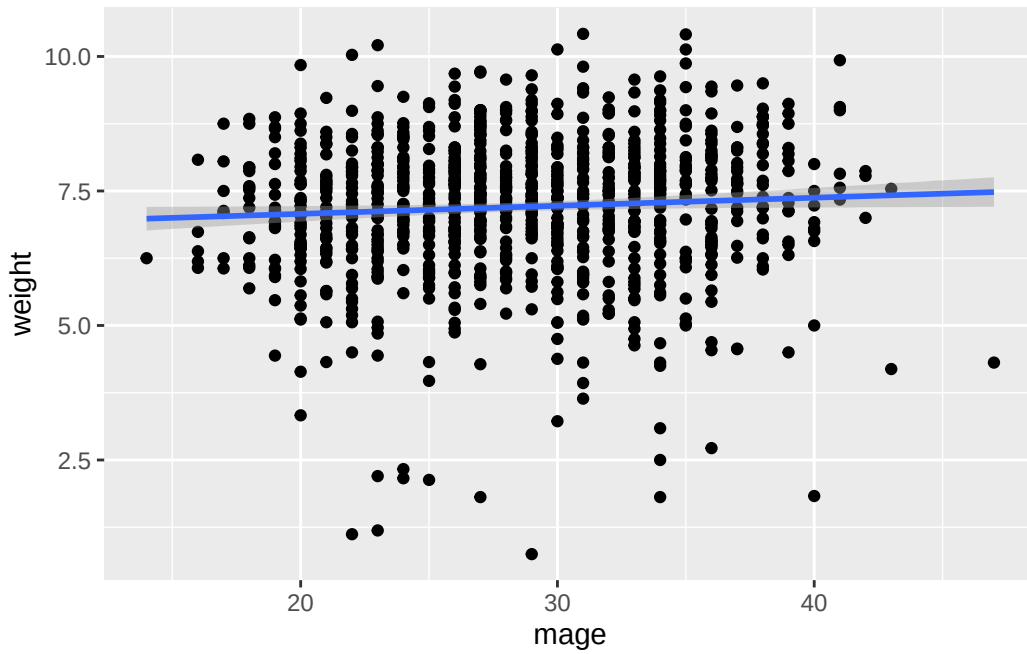
a. Another researcher is interested in assessing the relationship between babies' weights and mothers' ages. Fit another linear model to investigate this relationship. Provide the summary output below.

```r
#linear model
mage_model<- lm (weight ~ mage, data = births14_habitgiven)

#tidy summary
tidy(mage_model, conf.int = FALSE, conf.level = 0.95)
```

```
# A tibble: 2 x 5
  term        estimate std.error statistic   p.value
  <chr>          <dbl>     <dbl>     <dbl>     <dbl>
1 (Intercept)   6.78     0.208       32.6  2.57e-158
2 mage          0.0150   0.00717      2.09 3.70e-  2
```

```r
#add regression line
ggplot(births14_habitgiven,aes(x= mage, y = weight)) +
  geom_point()+
  geom_smooth(method = "lm")
```

```
`geom_smooth()` using formula = 'y ~ x'
```



b. In 2-3 sentences, explain how the regression line to model these data is fit, i.e., based on what criteria R determines the regression line.

The fit of a regression line is based on minimizing the difference between an observed data point for birth weight and the predicted birth weight for the same x value(mother's age).

c. Interpret the intercept in the context of the data and the research question. Is the intercept meaningful in this context? Why or why not?

Intercept is the estimated birth weight, when mother's age is 0. While not logical in the realistic context, the intercept is an extrapolation to help anchor the line of best fit equation rather that be used by itself.

d. Interpret the slope in the context of the data and the research question.

Slope is the value of increase in weight for every increase year of mother's age, showing by what factor of increase should the regression line increase. In this case, slope = 0.01498, shows that mother's age a very small but positive relationship with birth weight .

Part 2 - Parasites

Parasites can cause infectious disease – but not all animals are affected by the same parasites. Some parasites are present in a multitude of species and others are confined to a single host. It is hypothesized that closely related hosts are more likely to share the same parasites. More specifically, it is thought that closely related hosts will live in similar environments and have similar genetic makeup that coincides with optimal conditions for the same parasite to flourish.

In this part of the lab, we will see how much evolutionary history predicts parasite similarity.

The dataset comes from an Ecology Letters paper by Cooper at al. (2012) entitled "Phylogenetic host specificity and understanding parasite sharing in primates" located here. The goal of the paper was to identify the ability of evolutionary history and ecological traits to characterize parasite host specificity.

Each row of the data contains two species, `species1` and `species2`.

Subsequent columns describe metrics that compare the species:

- `divergence_time`: how many (millions) of years ago the two species diverged. i.e. how many million years ago they were the same species.
- `distance`: geodesic distance between species geographic range centroids (in kilometers)
- `BMdiff`: difference in body mass between the two species (in grams)
- `precdiff`: difference in mean annual precipitation across the two species geographic ranges (mm)
- `parsim`: a measure of parasite similarity (proportion of parasites shared between species, ranges from 0 to 1.)

The data are available in `parasites.csv` in your `data` folder.

Question 3

Let's start by reading in the `parasites` data and examining the relationship between `divergence_time` and `parsim`.

a. Load the data and save the data frame as `parasites`.

```
library(readr)
parasites <- read_csv("data/parasites.csv")
```

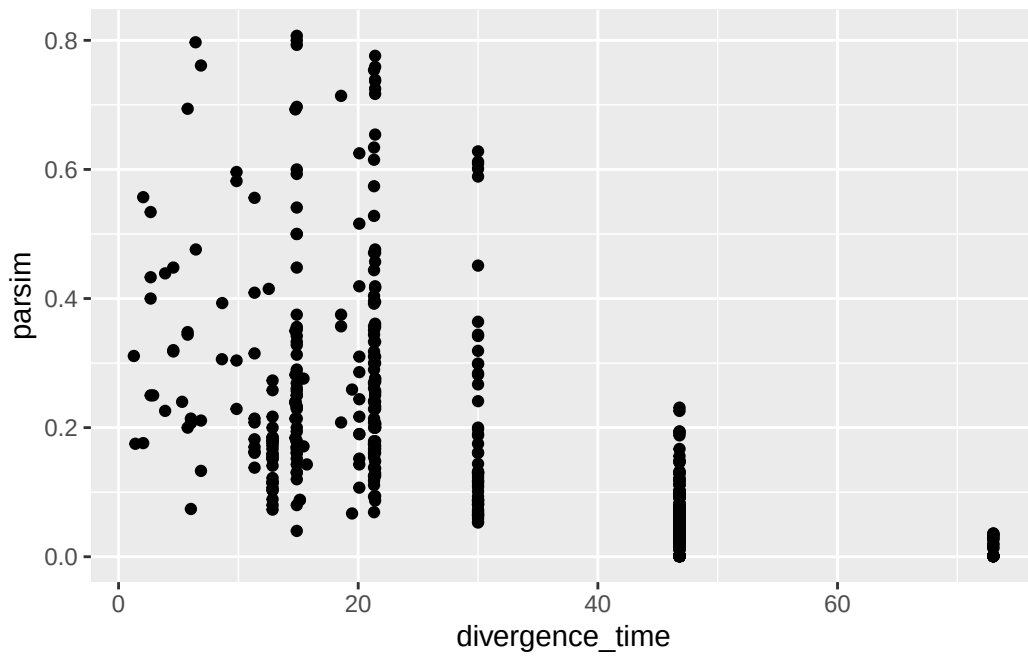
```
Rows: 595 Columns: 7
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (2): species1, species2
dbl (5): divergence_time, distance, BMdiff, precdiff, parsim

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

b. Based on the goals of the analysis, what is the response variable?

the `parsim`: a measure of parasite similarity

c. Visualize the relationship between the two variables.

```
ggplot(parasites, aes(x= divergence_time, y =parsim ))+
  geom_point()
```



d. Use the visualization to describe the relationship between the two variables.

There seems to be a strong negative relationship between divergence time and parasitism similarity between the 2 species. As time increase and the species evolve apart from one another in-terms of genetic and physical similarity, the measure of parasitic similarity has a decrease.

Question 4

Next, model this relationship.

a. Fit the model and write the estimated regression equation.

```
#linear model
parasim_model<- lm (parsim ~ divergence_time, data = parasites)

#tidy summary
tidy(parasim_model, conf.int = FALSE, conf.level = 0.95)
```

```
# A tibble: 2 x 5
  term            estimate std.error statistic   p.value
  <chr>              <dbl>     <dbl>     <dbl>     <dbl>
1 (Intercept)      0.379    0.0121        31.4 4.59e-128
2 divergence_time -0.00653  0.000312     -20.9 2.70e- 73
```

b. Interpret the slope and the intercept in the context of the data.

- **Slope = -0.0065**

  For each increase in million years, there is decrease in proportions of parasite similarity by 0.0065. This shows a weak negative relationship between divergence time and parasite similarity, therefore, increase difference between two species relates to decrease similarities in the parasites of the species themselves.

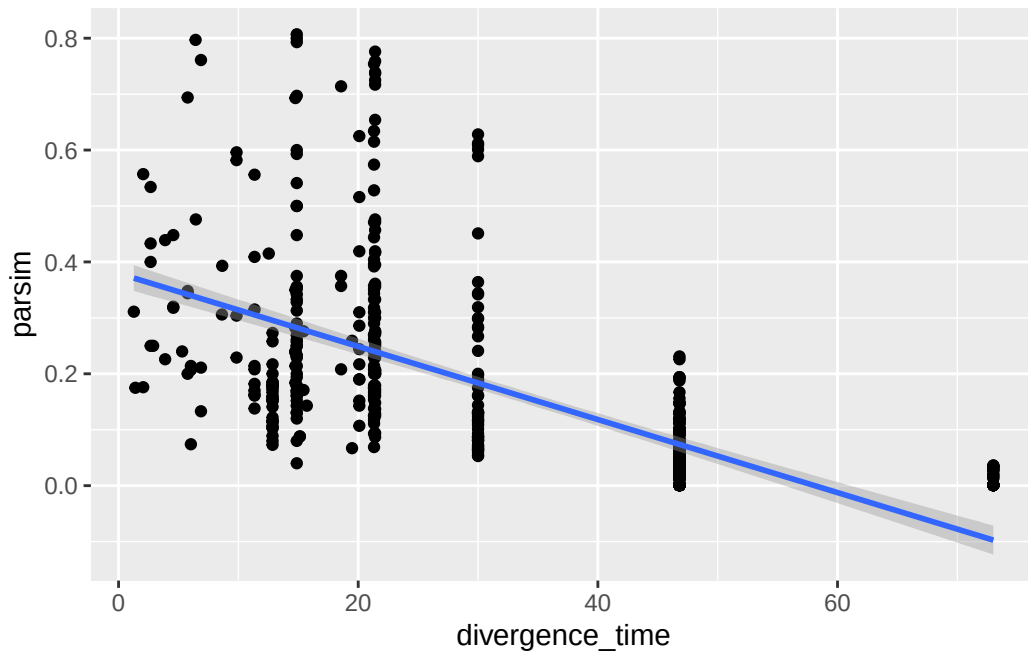
- **Intercept = 0.3795**

  At a point in time (0million years), where the two species was initally one species, the parasite similarity was 0.3795. This value is the maximum value for parasite similarity on average.

c. Recreate the visualization from Question 3, this time adding a regression line to the visualization.

```
#add regression line
ggplot(parasites, aes(x= divergence_time, y =parsim)) +
  geom_point()+
  geom_smooth(method = "lm")
```

```
`geom_smooth()` using formula = 'y ~ x'
```



d. What do you notice about the prediction (regression) line that may be strange, particularly for very large divergence times?

The regression line visualization shows a predicted parsim value out of the 0-1 range and into the negative values as divergence time increases signifcantly.

Question 5

Since `parsim` takes values between 0 and 1, we want to transform this variable so that it can range between $(-\infty, +\infty)$. This will be better suited for fitting a regression model (and interpreting predicted values!)

a. Using mutate, create a new variable `transformed_parsim` that is calculated as `log(parsim/(1-parsim))`. Add this variable to your data frame.

```
parasites <- parasites |>
  mutate(transformed_parsim = log(parsim / (1 - parsim)))
head(parasites)
```

```
# A tibble: 6 x 8
  species1        species2       divergence_time distance BMdiff precdiff parsim
  <chr>           <chr>                    <dbl>    <dbl>  <dbl>    <dbl>  <dbl>
1 Alouatta_caraya Alouatta_guar~            4.57     932.   389.     57.5   0.318
2 Alouatta_caraya Alouatta_pall~            3.89    3456.  1000.     54.9   0.226
3 Alouatta_caraya Alouatta_seni~            2.89    1908.   821.      9.53  0.25
```

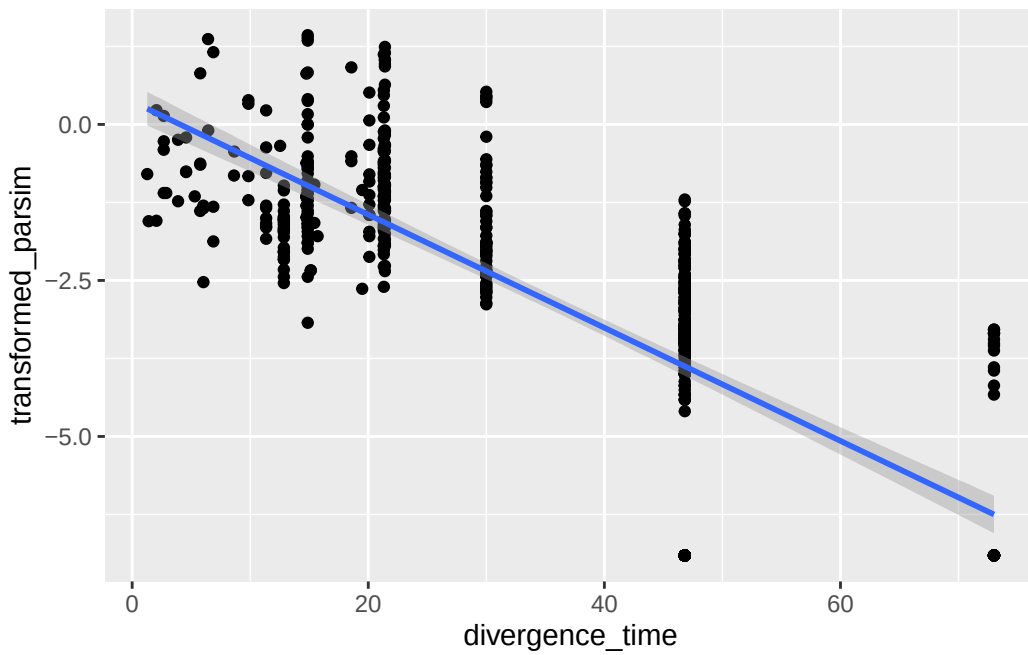
```
4 Alouatta_caraya Aotus_trivirg~          21.3     1497.  4665.    27.6   0.154
5 Alouatta_caraya Ateles_geoffr~          14.8     3785.  2005.    54.9   0.24
6 Alouatta_caraya Ateles_panisc~          14.8     1422.  3120.    16.7   0.214
# i 1 more variable: transformed_parsim <dbl>
```

> **Note**
>
> `log()` in R represents the **nautral log**.

b. Then, visualize the relationship between divergence_time and `transformed_parsim`. Add a regression line to your visualization.

```
#add regression line
ggplot(parasites, aes(x= divergence_time, y =transformed_parsim)) +
  geom_point()+
  geom_smooth(method = "lm")
```

```
`geom_smooth()` using formula = 'y ~ x'
```



c. Write a 1-2 sentence description of what you observe in the visualization.

After the transformation, the observed data is more linear and fiting with the regression line. Additionally, by adjusting the range, the predictive assuptions by the regression line makes more sense.

Question 6

Which variable is the strongest individual predictor of parasite similarity between species?

To answer this question, begin by fitting a linear regression model to each pair of variables. Do not report the model outputs in a tidy format but save each one as `dt_model`, `dist_model`, `BM_model`, and `prec_model`, respectively.

- `divergence_time` and `transformed_parsim`
- `distance` and `transformed_parsim`
- `BMdiff` and `transformed_parsim`
- `precdiff` and `transformed_parsim`

a. Report the slopes for each of these models. Use proper notation.

```r
dt_model<- lm (transformed_parsim ~ divergence_time, data = parasites)
dist_model<- lm (transformed_parsim ~ distance, data = parasites)
BM_model<- lm (transformed_parsim ~ BMdiff, data = parasites)
prec_model<- lm (transformed_parsim ~ precdiff, data = parasites)

#slope
coef(dt_model)[2]
```

```
divergence_time
    -0.09066296
```

```r
coef(dist_model)[2]
```

```
     distance
-0.0002739886
```

```r
coef(BM_model)[2]
```

```
       BMdiff
-5.044374e-06
```

```r
coef(prec_model)[2]
```

```
    precdiff
-0.006012838
```

divergence_time: Δy = -0.0907

distance: Δy = -0.0003

BMdiff: Δy = -5.0444e-06

precdiff: Δy = -0.0060

b. To answer the question of interest, would it be useful to compare the slopes in each model to choose the variable that is the strongest predictor of parasite similarity? Why or why not?

The domains of each variable pair are very different from one another (eg, divergence_time in million years vs distance in km). Thus, the values of the slopes alone are not comparable for any conclusions between the variables.

Question 7

Now, what if we calculated R^2 to help answer our question? To compare the explanatory power of each individual predictor, we will look at R^2 between the models. R^2 is a measure of how much of the variability in the response variable is explained by the model.

As you may have guessed from the name R^2 can be calculated by squaring the correlation when we have a simple linear regression model. The correlation r takes values -1 to 1, therefore, R^2 takes values 0 to 1. Intuitively, if r=1 or -1, then R^2=1, indicating the model is a perfect fit for the data. If r 0 then R^2 0, indicating the model is a very bad fit for the data.

You can calculate R^2 using the glance function. For example, you can calculate R^2 for `dt_model` using the code `glance(dt_model)$r.squared`.

a. Calculate and report R^2 for each model fit in the previous exercise.

```
print("divergence_time")
```

```
[1] "divergence_time"
```

```
glance(dt_model)$r.squared
```

```
[1] 0.5125147
```

```
print("distance")
```

```
[1] "distance"
```

```
glance(dist_model)$r.squared
```

```
[1] 0.4432029
```

```r
print("BMdiff")
```

```
[1] "BMdiff"
```

```r
glance(BM_model)$r.squared
```

```
[1] 0.003269571
```

```r
print("precdiff")
```

```
[1] "precdiff"
```

```r
glance(prec_model)$r.squared
```

```
[1] 0.008611603
```

b. To answer our question of interest, would it be useful to compare the R^2 in each model to choose the variable that is the strongest predictor of parasite similarity? Why or why not?

   Because R^2 is the coefficient of determination, it shows the proportion of variance in the specific variable comparison in a scale-free method. Through this, each value of R^2 is isolated in its analysis but comparable in representing the fit of the model. Thus, divergence_time has the strongest predictor of parasite similarity, and BMdiff is the weakest predictor.

Part 3 - GDP and life expectancy

Gapminder is a "fact tank" that uses publicly available world data to produce data visualizations and teaching resources on global development. We will use an excerpt of their data to explore relationships among world health metrics across countries and regions between the years 1952 and 2007. The data set is called `gapminder`, from the **gapminder** package. A table of variables can be found below.

- `country`: The country name
- `continent`: The continent name
- `year`: ranges from 1952 to 2007 in increments of 5 years
- `lifeExp`: life expectancy at birth, in years
- `pop`: population of country
- `gdpPercap`: GDP per capita (US$, inflation-adjusted)

Question 8

a. **Data:** For our analysis, we will only be working with data from 2007. Below, filter the data set so only values from the year 2007 are shown. Save this data set as `gapminder_07` and use it for the remainder of this exercise and the following.

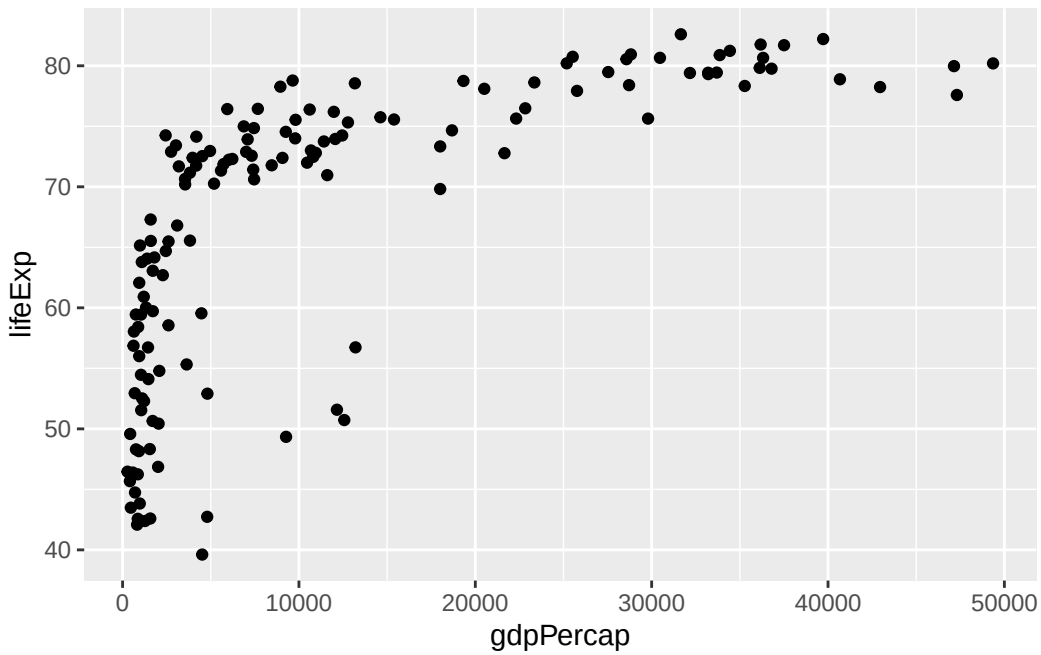
```
library(gapminder)
gapminder_07 <- gapminder |>
  filter(year ==2007)

head(gapminder_07)
```

```
# A tibble: 6 x 6
  country     continent  year lifeExp      pop gdpPercap
  <fct>       <fct>     <int>   <dbl>    <int>     <dbl>
1 Afghanistan Asia       2007    43.8 31889923      975.
2 Albania     Europe     2007    76.4  3600523     5937.
3 Algeria     Africa     2007    72.3 33333216     6223.
4 Angola      Africa     2007    42.7 12420476     4797.
5 Argentina   Americas   2007    75.3 40301927    12779.
6 Australia   Oceania    2007    81.2 20434176    34435.
```

b. **Visualization:** We are interested in learning more about GDP, and we'll start with exploring the relationship between life expectancy and GDP. Create two visualizations:

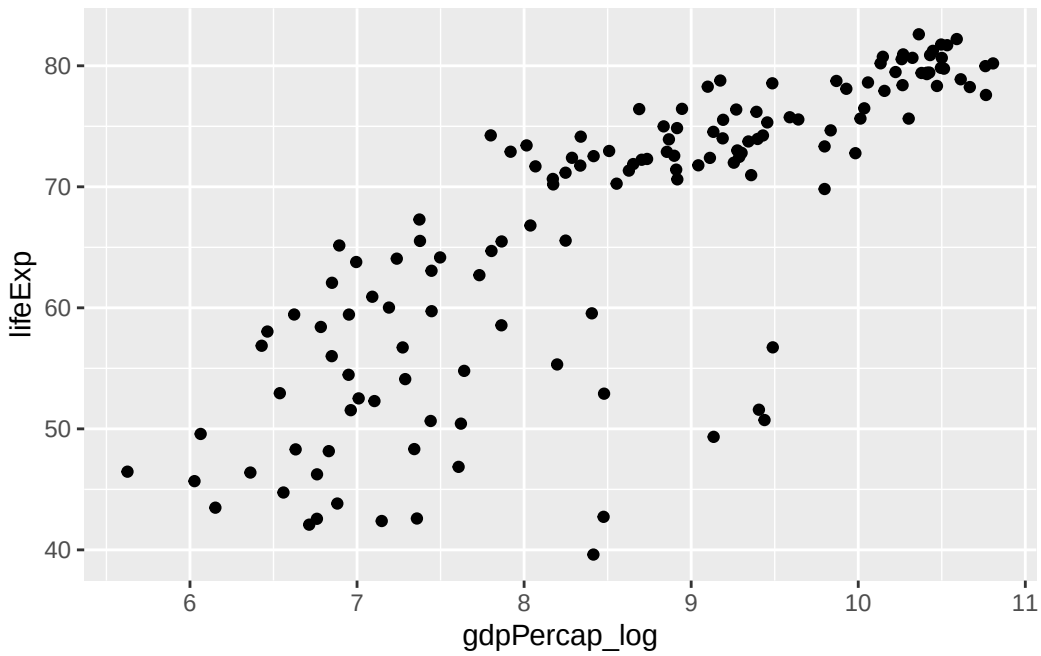
- Scatter plot of `gdpPercap` vs. `lifeExp`

```
ggplot(gapminder_07, aes(x = gdpPercap, y = lifeExp)) +
  geom_point()
```



- Scatter plot of `gdpPercap_log` vs. `lifeExp`, where `gdpPercap_log` is a new variable you add to the data set by taking the natural log of `gdpPercap`.

```
gapminder_07 <- gapminder_07 |>
  mutate(gdpPercap_log = log(gdpPercap))

#graph
ggplot(gapminder_07, aes(x = gdpPercap_log, y = lifeExp)) +
  geom_point()
```



First describe the relationship between each pair of the variables. Then, comment on which relationship would be better modeled using a linear model, and explain your reasoning.

There is a strong positive relationship between life expectancy and GDP per capita, which is better shown in the second model (using log), as it is a linearized version of the raw data and better fit for a linear regression application.

c. **Model fitting:**

- Fit a linear model predicting log gross domestic product from life expectancy. Display the tidy summary.

```
#linear model
life_expectancy_model<- lm (lifeExp ~ gdpPercap_log, data = gapminder_07)

#tidy summary
tidy(life_expectancy_model, conf.int = FALSE, conf.level = 0.95)
```

```
# A tibble: 2 x 5
  term          estimate std.error statistic  p.value
  <chr>            <dbl>     <dbl>     <dbl>    <dbl>
1 (Intercept)       4.95     3.86       1.28 2.02e- 1
2 gdpPercap_log     7.20     0.442     16.3  4.12e-34
```

d. **Model evaluation:**

- Calculate the R-squared of the model using two methods and confirm that the values match: first method is using `glance()` and the other method is based on the value of the correlation coefficient between the two variables.

```
print("Glance Method:")
```

```
[1] "Glance Method:"
```

```
glance(prec_model)$r.squared
```

```
[1] 0.008611603
```

```
print("Correlation Method:")
```

```
[1] "Correlation Method:"
```

```
corr_r_squared <- cor(gapminder_07$lifeExp, gapminder_07$gdpPercap_log)^2
corr_r_squared
```

```
[1] 0.654449
```

- Interpret R-squared in the context of the data and the research question.

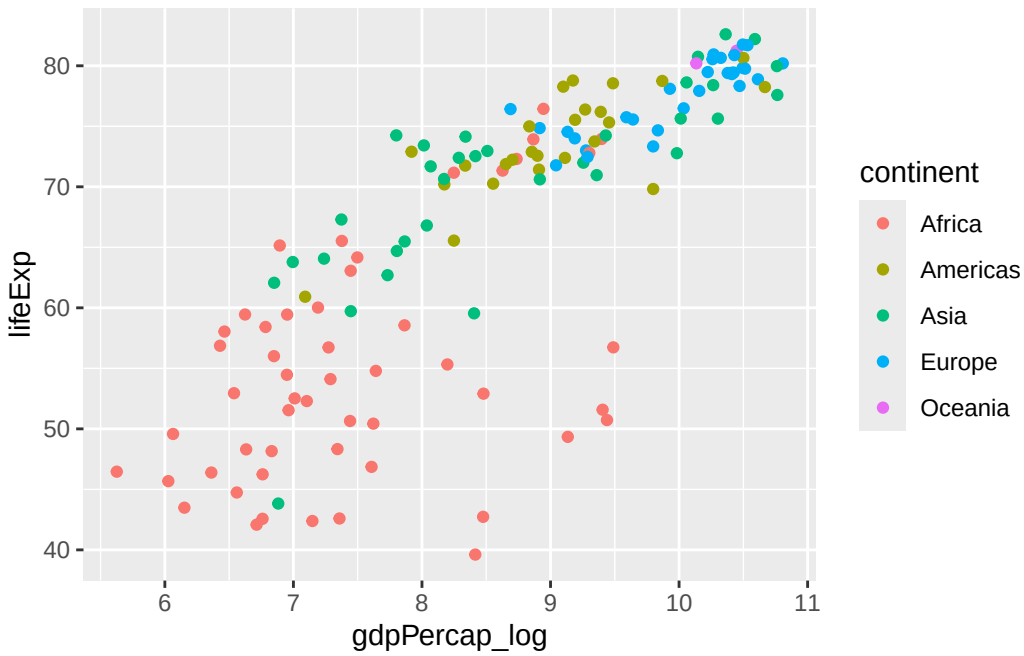
The R-squared value indicates that about 65 .4% of the variation in life expectancy across countries in 2007 can be explained by differences in GDP per capita. This suggests a strong positive association between the varibles. However, ~30% of the variation remains unexplained by the model, which could be attributed to other factors.

Question 9

Next, we want to examine if the relationship between GDP and life expectancy that we observed in the previous exercise holds across all continents in our data. We'll continue to work with logged GDP (`gdpPercap_log`) and data from 2007.

a. **Justification:** Create a scatter plot of `gdpPercap_log` vs. `lifeExp`, where the points are colored by `continent`. Do you think the trend between `gdpPercap_log` and `lifeExp` is different for different continents? Justify your answer with specific features of the plot.

```
ggplot(gapminder_07, aes(x = gdpPercap_log, y = lifeExp, colour = continent)) +
  geom_point()
```



Yes, the trend appears different across continents based on the scatter plot. For example, Africa shows the widest spread with generally lower GDP and lower life expectancy. The slope appears flatter with many countries below the global trend. On the other hand, Europe & Oceania show the strongest positive relationship with consistently high life expectancy (75-82 years) across a range of GDP values

a. **Model fitting and interpretation:**

- Regardless of your answer in part (a), fit an additive model (main effects) that predicts `gdpPercap_log` from life expectancy and continent (with Africas as the baseline level). Display a tidy summary of the model output.

```r
continent_model <- lm(lifeExp ~ gdpPercap_log + continent, data = gapminder_07)

tidy(continent_model, conf.int = FALSE, conf.level = 0.95)
```

```
# A tibble: 6 x 5
  term              estimate std.error statistic  p.value
  <chr>                <dbl>     <dbl>     <dbl>    <dbl>
1 (Intercept)          20.1      4.03       4.99 1.79e- 6
2 gdpPercap_log         4.63     0.527      8.78 6.14e-15
3 continentAmericas    11.7      1.65       7.07 7.46e-11
4 continentAsia        10.1      1.48       6.85 2.31e-10
5 continentEurope      11.3      1.89       5.95 2.14e- 8
6 continentOceania     12.9      4.52       2.86 4.91e- 3
```

- Interpret the *intercept* of the model, making sure that your interpretation is in the units of the original data (not on log scale).

When GDP per capita is $1 (log = 0), the model predicts a life expectancy of approximately ~20 years for African countries. However, this is not meaningful in practice because no country has a GDP per capita of $1, and this represents extreme extrapolation outside the range of observed data.

- Interpret the *slope* of the model, making sure that your interpretation is in the units of the original data (not on log scale).

The slope for `gdpPercap_log` represents the expected change in life expectancy (in years) for a one-unit increase in log(GDP per capita)

b. **Prediction:** Predict the GDP of a country in Asia where the average life expectancy is 70 years old.

Question 10

Communication is a critical yet often overlooked part of data science. When we engage with our audience and capture their interest, we can ultimately better communicate what we are trying to share.

Please watch the following video: Hans Rosling: 200 years in 4 minutes.

Then, write a paragraph (4-5 sentences) addressing the following:

- What did you enjoy about the presentation of data? What did you find interesting
- Were there any aspects of the presentation that were hard to follow? If so, what?
- What are your general take-aways from this presentation?
- What are your general take-aways from how this presentation was given?

The presentation transforms global development data through animated storytelling which maintains audience interest throughout the entire show. The moving bubble charts showed how countries evolved from their original state between two hundred years and helped understanding of complex statistical ideas through visual demonstrations. The most interesting part showed how China and Japan developed as they followed Western countries because both nations showed progress in health and wealth which disproved existing beliefs about worldwide inequality. The audience found it difficult to track country-specific movements during the first half of the 1800s because bubbles moved quickly but Rosling's physical demonstrations created an engaging and understandable experience. Global health and wealth have improved throughout history because the world has experienced unprecedented

progress which proves people wrong who share pessimistic views about development. The storytelling elements together with energetic performance and data visualizations function as essential components for effective data communication which enables audience members to emotionally connect with data points.

Wrap-up

Submission

Once you are finished with the lab, you will submit your final PDF document to Canvas.

> ⚠ Warning
>
> Before you wrap up the assignment, make sure all of your documents are updated on your GitHub repo. We will be checking these to make sure you have been practicing how to commit and push changes.
> You must turn in a PDF file to the Canvas page by the submission deadline to be considered "on time".

To submit your assignment:

- Go to http://www.Canvas.com and click *Log in* in the top right corner.
- Click *School Credentials* → *Duke NetID* and log in using your NetID credentials.
- Click on your *REN R 213* course.
- Click on the assignment, and you'll be prompted to submit it.
- Mark all the pages associated with question. All the pages of your lab should be associated with at least one question (i.e., should be "checked").

> ❗ Checklist
>
> Make sure you have:
>
> - attempted all questions
> - rendered your Quarto document
> - committed and pushed everything to your GitHub repository such that the Git pane in RStudio is empty
> - uploaded your PDF to Canvas
> - selected pages associated with each question on Canvas

Grading

The lab is graded out of a total of 4 points.

You can earn up to 4 points on each question:

- 4: Response shows excellent understanding and addresses all or almost all of the rubric items.
- 3: Response shows good understanding and addresses most of the rubric items.
- 2: Response shows understanding and addresses a majority of the rubric items.
- 1: Response shows effort and misses many of the rubric items.
- 0: Response does not show sufficient effort or understanding and/or is largely incomplete.